

**WEATHER: RMSE at each forecast step**  
**Lookback = 24 steps, Forecast horizon = 24 steps**

| Forecast step | CNN    | LSTM   | dCeNN+ELM | dCeNN+ELM+ASP |
|---------------|--------|--------|-----------|---------------|
| 1.000         | 23.973 | 30.401 | 26.294    | 24.746        |
| 2.000         | 28.631 | 29.422 | 30.069    | 28.849        |
| 3.000         | 31.834 | 31.018 | 32.464    | 31.441        |
| 4.000         | 34.050 | 32.249 | 34.246    | 33.322        |
| 5.000         | 35.145 | 33.198 | 35.702    | 34.799        |
| 6.000         | 35.934 | 34.025 | 36.929    | 36.020        |
| 7.000         | 36.872 | 34.690 | 37.523    | 36.582        |
| 8.000         | 37.525 | 35.271 | 37.867    | 36.878        |
| 9.000         | 38.120 | 35.848 | 38.083    | 37.078        |
| 10.000        | 38.703 | 36.439 | 38.309    | 37.292        |
| 11.000        | 38.707 | 36.915 | 38.553    | 37.488        |
| 12.000        | 38.911 | 37.350 | 38.634    | 37.528        |
| 13.000        | 39.106 | 37.710 | 38.734    | 37.645        |
| 14.000        | 39.486 | 38.101 | 38.882    | 37.812        |
| 15.000        | 39.613 | 38.527 | 38.953    | 37.843        |
| 16.000        | 39.552 | 38.984 | 38.911    | 37.747        |
| 17.000        | 39.487 | 39.371 | 38.898    | 37.712        |
| 18.000        | 39.724 | 39.622 | 38.937    | 37.753        |
| 19.000        | 39.835 | 39.726 | 39.105    | 37.966        |
| 20.000        | 39.059 | 39.702 | 39.429    | 38.315        |
| 21.000        | 38.993 | 39.589 | 39.658    | 38.584        |
| 22.000        | 39.008 | 39.473 | 39.784    | 38.707        |
| 23.000        | 39.147 | 39.507 | 39.903    | 38.758        |
| 24.000        | 38.982 | 39.861 | 40.087    | 38.828        |

• This table shows how prediction error changes from the first forecast step to the last forecast step.  
• It is especially useful for explaining whether a model degrades smoothly or sharply as it predicts further into the future.  
• Lower is better. Large mistakes are penalized more strongly than in MAE.  
• Weather combines multiple physical variables. Aggregate RMSE or MAE should therefore be read mainly as a model-comparison tool, not as a single physical-unit error.